A uniform Ricci-energy gap above the classical Einstein envelope

23 September 2026. This is an application of the parameterized functional theorem, with actual polarized monopole classes. The elementary energy envelope below follows from classical inequalities. The added conclusion is a uniform positive separation from that envelope, persisting under integral Einstein error. No new topological nonexistence theorem is asserted.

1. Statement on a realized family

Let

$$M = X\#k\overline{\mathbb{CP}}^2\#m(S^1\times S^3), \qquad A = c_1^2(X) > 0, \quad h = \frac{k+4m}{A}, \quad 0 < h < \frac{1}{3}, \tag{1}$$

where X is a minimal complex surface of general type. Put

$$R = \int_M |\operatorname{Ric}|^2, \qquad e = \frac{R}{8\pi^2 A}, \qquad F(h) = \frac{(4-\sqrt{1-3h})^2}{9},$$

$$\mathscr{E}(g) = \inf_{\lambda\neq 0} \frac{\int_M |\operatorname{Ric}-\lambda g|^2}{\lambda^2 \operatorname{Vol}(g)}. \tag{2}$$

Both e and $\mathscr{E}$ are invariant under constant rescaling.

Theorem. Relative to Theorem 1 of the functional-pencil proof, there is a universal $\varepsilon > 0$, independent of X, k, m, h, g, such that

$$\boxed{\mathscr{E}(g) + (e - F(h))_+ \geqslant \varepsilon h^2.} \tag{3}$$

Thus, for example, $\mathscr{E}(g) \leqslant \varepsilon h^2/2$ implies $e \geqslant F(h) + \varepsilon h^2/2$. Every Einstein metric on M, if one exists, necessarily satisfies

$$\boxed{\frac{R}{8\pi^2 A} \geqslant F(h) + \varepsilon h^2.} \tag{4}$$

An actual nonzero monopole projection excludes Einstein constants zero and positive, so (4) concerns all Einstein candidates, not just a chosen sign.

The same theorem holds whenever $T = 2\chi+3\tau = A(1-h)$ and each metric admits an actual conformally solvable Spin^c structure with $p = (a_g^+)^2 \geqslant A$ and $a^2 \geqslant T$. No regularity of the class selection in g is needed. In (1), the polarized sign selection gives

$$p \geqslant A, \qquad a^2 = A - k \geqslant A - k - 4m = T. \tag{5}$$

This is gauge-theoretic input, not just an intersection-form calculation; the conformal class keeps its polarization and its selected structure. For $b_+ = 1$, the small-perturbation chamber and, for handles, the loop insertions are those in the polarized closure. The source is LeBrun, Proposition 3.2 and its proof.

For a concrete family without chamber issues, take $X = \Sigma_g \times \Sigma_q$, $g, q \geqslant 2$. Then $A = 8(g-1)(q-1)$ and $b_+ > 1$. In particular, $(\Sigma_2\times\Sigma_2)\#2\overline{\mathbb{CP}}^2$ has $A = 8, h = 1/4$. Formula (3) applies to every smooth metric on this actual closed manifold. Any Einstein metric on it would satisfy

$$\frac{R}{64\pi^2} \geqslant \frac{49}{36} + \frac{\varepsilon}{16}.$$

We do not assume or produce an Einstein metric on it.

2. The classical envelope is not the new result

For an Einstein metric, set

$$x^2 = \frac{\int s^2}{8\pi^2 A}, \qquad y^2 = \frac{6\int |W^+|^2}{8\pi^2 A}.$$

The classical scalar and scalar–Weyl inequalities and Chern–Weil give

$$x \geqslant 2, \qquad x + y \geqslant 3, \qquad x^2 + 8y^2 = 12(1-h), \qquad e = x^2/4. \tag{6}$$

These imply $x \geqslant (8 - 2\sqrt{1-3h})/3$, hence $e \geqslant F(h)$. Equality is feasible in this finite relaxation; it is not claimed to be realized by a metric. The constraints are the scalar estimate, LeBrun's Theorem 2.3, and the exact Chern–Weil identity. This optimization is entirely classical.

The all-metrics infimum in (1) is $\inf e = 1 + h$, by LeBrun's Theorem 4.3. One has the exact comparison

$$F(h) - (1+h) = \frac{4h^2}{(1+\sqrt{1-3h})^2} > 0. \tag{7}$$

The new term in (4) is therefore an additional uniform energy separation above the already optimized classical restriction, not the separation in (7). Classical equality rigidity already excludes exact Einstein saturation away from the Kähler–Einstein endpoint; it does not alone give a uniform gap for varying manifolds or almost-Einstein metrics.

3. Exact Chern–Weil reduction and the proof

Use tensor Ricci norms and Hilbert–Schmidt Weyl operator norms. Write

$$P = \int |\mathrm{Ric}^\circ|^2, \quad S_2 = \int s^2, \quad W_2 = \int |W^+|^2.$$

The identities

$$S_2 = 4(R - P), \qquad W_2 = 2\pi^2 T - R/12 + P/3$$

turn the functional in the companion theorem into

$$\mathfrak{D}_\beta = K_\beta R + L_\beta P + 12\pi^2(\beta+1)T - 72\pi^2 p, \tag{8}$$

where

$$K_\beta = \frac{(\beta+1)(8-\beta)}{2\beta}, \qquad L_\beta = \frac{2(\beta+1)(\beta-2)}{\beta}.$$

Let $z = \sqrt{1-3h}$ and choose

$$\beta = \beta(h) = \frac{8-2z}{1+2z} \in (2, 8). \tag{9}$$

Direct substitution gives

$$K_\beta F(h) + \tfrac{3}{2}(\beta+1)(1-h) - 9 = 0.$$

Using (5) in (8), with no discarded negative error, yields

$$\mathfrak{D}_\beta = 8\pi^2 A K_\beta (e - F(h)) + L_\beta P - 72\pi^2(p - A) \leqslant 8\pi^2 A K_\beta (e - F(h)) + L_\beta P. \tag{10}$$

First fix the Einstein target to $-g$, and let $V = \mathrm{Vol}(g)$, $r = V^{-1}\int |\mathrm{Ric} + g|^2$. The functional-pencil theorem, with $L = 8$ and $a^2 \geqslant T$, supplies a universal $c > 0$ such that

$$c(\beta - 2)^2 \leqslant r + \mathfrak{D}_\beta / V.$$

Since $P/V \leqslant r$, equation (10) implies

$$c(\beta-2)^2 \leqslant (1+L_\beta)r + K_\beta \frac{8\pi^2 A}{V}(e-F(h)). \tag{11}$$

The scalar monopole bound and the trace inequality give

$$32\pi^2 A \leqslant S_2, \qquad \sqrt{S_2/V} \leqslant 4 + 2\sqrt{r}, \qquad \frac{8\pi^2 A}{V} \leqslant (2+\sqrt{r})^2. \tag{12}$$

For $r \leqslant 1$, replace the last factor in (11) by its positive part before using (12). On $[2,8]$,

$$0 \leqslant K_\beta \leqslant 9/2, \qquad 0 \leqslant L_\beta \leqslant 27/2.$$

Consequently

$$c(\beta-2)^2 \leqslant \frac{81}{2}\Big[r + (e-F(h))_+\Big]. \tag{13}$$

Finally,

$$\beta(h) - 2 = \frac{18h}{(1+z)(1+2z)} \geqslant 3h.$$

Thus (3), with r in place of $\mathscr{E}$, holds for $r \leqslant 1$ with coefficient $2c/9$. For $r > 1$ it is immediate after choosing $\varepsilon = \min\{2c/9, 1\}$, because $h^2 < 1/9$. For any $\lambda < 0$, apply this result to $(-\lambda)g$: its normalized error is the corresponding quotient in (2), while e and h are unchanged. Taking the infimum proves (3) first for $\mathscr{E}_-$, the infimum restricted to negative λ.

To include positive target constants, put $C = 8\pi^2 A$, $I = \int s$, and let $\mathscr{E}_+$ be the infimum for $\lambda > 0$. The signed scalar estimate is

$$\int s_-^2 \geqslant 32\pi^2 p \geqslant 4C, \qquad s_- := \max\{-s, 0\}. \tag{14}$$

For clarity, it follows directly from an ordinary monopole solution: with $\rho = |\Phi|^2$, the integrated spinor identity gives $\int \rho^2 \leqslant \int s_- \rho$, while Hodge projection gives $\int \rho^2 \geqslant 32\pi^2 p$; apply Cauchy–Schwarz. If $I > 0$, then

$$\frac{I^2}{V} \leqslant \int s_+^2 \leqslant S_2 - 4C \leqslant 4R - 4C, \qquad \mathscr{E}_+ = 4 - \frac{I^2}{RV} \geqslant \frac{4}{e}.$$

If $I \leqslant 0$, direct minimization gives $\mathscr{E}_+ = 4$. When $e \leqslant F(h) + 1 \leqslant 25/9$, therefore, $\mathscr{E}_+ \geqslant 36/25 > 1$. When $e > F(h) + 1$, the energy excess alone exceeds one. Hence $\mathscr{E}_+ + (e - F(h))_+ \geqslant 1 \geqslant \varepsilon h^2$. Together with $\mathscr{E} = \min\{\mathscr{E}_-, \mathscr{E}_+\}$, this proves (3) for all target signs.

Only the functional-pencil theorem is the nonclassical analytic input. There is no geometric compactness, scalar band, auxiliary-form assumption, or additional square-cost budget in this transfer.

4. Why the parameter eight does not give this energy separation

At $\beta = 8$, $K_8 = 0$. Equation (8) loses R altogether:

$$\mathfrak{D}_8 = 108\pi^2(T - 2p/3) + 27P/2. \tag{15}$$

This is the old topological gap. Varying β retains the independent energy coordinate and gives the separation in (3).

There is an exact algebraic non-implication check. At $h = 1/4$, set $p = A$, $a^2 = T = 3A/4$, $P = 0$, $s = -4$, and prescribe the constant Weyl norm $w = \sqrt{6}|W^+| = 8/7$. Choose

$$V = 2\pi^2 A(49/36), \qquad R = 4V.$$

These moment data satisfy Chern–Weil, the scalar inequality, every classical bound $\mathfrak{D}_\gamma \geqslant 0$, $\gamma > 0$, and $e = F(1/4) = 49/36$. However,

$$\mathfrak{D}_{7/2} = 0, \qquad \mathfrak{D}_8/V = 162/49. \tag{16}$$

They violate the new parameter-7/2 estimate. They do not violate any possible universal coefficient in the global parameter-eight member $r + \mathfrak{D}_8/V \geqslant 36c_8$: necessarily $c_8 \leqslant 1/11$. Indeed, for an actual negative Kähler–Einstein metric g_0, the rescalings $g = tg_0$ give

$$r + \mathfrak{D}_8/V = 4(1 - 1/t)^2 + 18/t^2 \geqslant 36/11,$$

with equality at $t = 11/2$. Yet $162/49 > 36/11$. This proves independence from the parameter-eight numerical inequality even at the largest coefficient it could possibly have.

The data in (16) are a formal curvature-moment model, not a metric construction. The manifold and its actual monopole-class selection in Section 1 are realized; realizing the forbidden moments would contradict the theorem. This argument does not claim independence from every possible use of the old differential proof machinery.

5. The remaining topological input is precise

Equation (3) does not exclude all energies on any new smooth topology. For normalized Einstein metrics, the missing independent number is

$$e = \frac{R}{8\pi^2 A} = \frac{V}{2\pi^2 A}. \tag{17}$$

It is not fixed by T, A, or p. Chern–Weil fixes a linear combination of scalar and Weyl energies, not this scalar/volume coordinate. An independently proved upper bound $e < F(h) + \varepsilon h^2$ for every Einstein candidate would convert (4) into a new topological obstruction; no such bound is supplied here. An all-metrics energy *infimum* cannot serve as that upper bound.

For the pointwise negative Ricci input $\delta_- = \| \operatorname{Ric}^\sharp + I \|_{\infty,\mathrm{op}}$, (3) implies the raw consequence

$$4\delta_-^2 + (e - F(h))_+ \geqslant \varepsilon h^2. \tag{18}$$

Thus a construction with controlled energy below the strengthened envelope cannot be arbitrarily Ricci-pinched. This is a restriction on such input metrics, not a proof that their smooth manifold has no Einstein metric at a different energy.

`check_line1_energy_envelope_gap.py` checks the exact identities, envelope optimization, constants, scale transformation, and formal non-implication witness. These finite checks do not certify Seiberg–Witten solvability or the inherited differential theorem. No priority claim over unexamined later literature is made.